Lighthouse prompt

64-bit RISC-V compute for XRBench-class real-time mobile XR
Vector-capable CPU, accelerator, or SoC block
3 W TDP target in a 3 nm-class low-power process
Report the tested candidates and their trade-offs.

Eight coupled domains: choices and constraints

XRBench workload
Multi-model streams, deadlines

ISA and ABI contract
Baseline, extensions, compatibility

CPU / accelerator / SoC block
Partitioning, vectors, interfaces

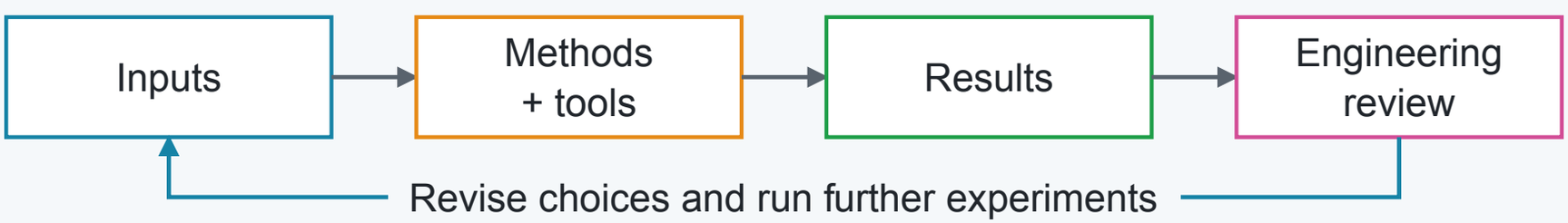
Memory and data movement
Locality, hierarchy, interconnect energy

3 W power envelope
Activity, sustained power, temperature

Compiler/runtime stack
Code generation, libraries, deployment

3 nm-class physical constraints
Process, timing, leakage, signoff

Reliability and verification
Correctness, corner cases, rejection

Evaluate the choices

Simulate workloads, implement candidates, and check correctness.